

INFRASTRUCTURE RESOURCE REPORT — PHASE 3

Infrastructure Report — Phase 3

Full Social Platform

v2.0 · Richard Uzor — May 2026 · Devstrike Digital Limited

01 — PHASE 3 INFRASTRUCTURE FOCUS

Phase 3 introduces the most significant infrastructure addition: a real-time WebSocket layer for fellowship rooms, prayer rooms, and DMs. A read replica is added to handle the read-heavy browse load. AI embedding generation is added for semantic search.

02 — NEW PHASE 3 COMPONENTS

WebSocket Server *Ably (managed) or Soketi (self-hosted on Railway)*

Tier: Ably: usage-based | Soketi: ~\$10/month

Estimated Cost: ~~₦32,000~~–~~₦64,000~~/month

Powers real-time messaging in fellowship rooms, live prayer sessions, and DMs. Ably recommended for Phase 3 launch — managed service, Nigerian PoPs, no DevOps overhead. Evaluate Soketi at Phase 3 maturity for cost optimisation.

Scaling trigger: Ably free tier: 200 concurrent connections, 6M messages/month. Upgrade when: concurrent connections regularly > 160 (80% of limit).

Database Read Replica *Railway PostgreSQL replica*

Tier: ~\$15–20/month

Estimated Cost: ₦24,000–₦32,000/month

All read-only API queries (GET endpoints) routed to replica. Write queries continue to primary. Dramatically improves browse-heavy load performance as archive grows to 250,000+ messages.

Scaling trigger: Add replica when: read:write ratio > 10:1, or GET query P95 consistently > 100ms.

AI Embedding Service *OpenAI text-embedding-3-small API*

Tier: Usage-based (~\$0.02/1M tokens)

Estimated Cost: ₦32,000–₦64,000/month

Generates 1536-dimension vector embeddings for every new message post. Stored in pgvector column. Powers semantic search — 'find everything said about healing by Yoruba pastors'. Cost is negligible per message (~₦0.05 per embedding).

Scaling trigger: Cost scales linearly with message volume. At 250,000 messages: total embedding cost ~₦2,500 one-time. Monthly cost is ongoing for new posts only.

Upstash Redis (paid) *Upstash Redis Pay-per-use*

Tier: Usage-based

Estimated Cost: ₦16,000–₦32,000/month

Now critical for: WebSocket presence caching, live prayer room session state, feed personalisation cache, premium subscription status cache. Free tier was sufficient for Phase 2 — Phase 3 social features push beyond it.

Scaling trigger: Upgrade from free tier when daily Redis commands consistently > 8,000.

03 — PHASE 3 FULL COST SUMMARY

Item	Monthly Cost
Vercel Pro Team	₦80,000
Railway (API + DB primary + replica + workers)	₦96,000–₦120,000

Item	Monthly Cost
WebSocket (Abyl)	₦32,000–₦64,000
Cloudflare R2 + CDN Pro	₦56,000–₦80,000
Upstash Redis	₦16,000–₦32,000
Resend (Business)	₦64,000–₦96,000
OpenAI embeddings	₦32,000–₦64,000
Sentry Team	₦42,000
Plausible Business	₦30,000
FCM	₦0
Misc + buffer (10%)	₦45,000–₦65,000
TOTAL	₦493,000–₦701,000/month

Cost vs Revenue

Phase 3 infrastructure cost of ~₦600,000/month against projected revenue of ₦4,000,000+ represents a 15% cost ratio — well within healthy SaaS margins. Infrastructure is not the constraint to growth at this stage.

04 — PHASE 3 DISASTER RECOVERY

Item	Phase 3 Policy
Database backups	Daily automated + point-in-time recovery (PITR) via Railway. 30-day retention.
RTO	1 hour — primary DB failover to replica in < 5 minutes, full restore from backup in < 1 hour.
RPO	1 hour — WAL streaming from primary to replica. Maximum 1 hour of data loss in worst case.
Room message history	PostgreSQL primary (replicated). Messages older than 12 months archived to R2 cold storage.
WebSocket failover	

Item	Phase 3 Policy
	Ably handles redundancy automatically. Soketi: deploy 2 instances with load balancer.
Incident response	Page on-call engineer immediately. Status page updated within 15 minutes. Post-mortem within 24 hours.